One Label, Many Records: Shared Prediction Error in Regression

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Abstract

Researchers often reuse one predicted item label across many records and construct several aggregates from those records. The resulting errors can enter regression outcomes, covariates, and several estimates at once. I represent reuse by item-to-record loading matrices and derive its effects on coefficient bias and covariance. An outcome-regression coefficient is linear in item errors; a constructed covariate also changes the regression's quadratic denominator. Independent errors on distinct items can therefore produce dependent errors across records, while shared errors across constructed variables alter their cross-products. I relate these consequences to existing generated-regressor, measurement-error, and shared-source inference. For an existing probability audit of items, I give joint regression-moment corrections and an estimable covariance, distinguishing their exact design properties from the conditions required for normal coefficient intervals. Simulations assess bias, estimated SEs, and coverage; normal intervals can under-cover when item influence is concentrated. An illustration using a June 2022 YouGov browsing sample imposes classification errors on observed visit patterns. A separate comparison uses observed adult-domain classifications. These examples distinguish error counts, error locations, and the validation information needed for a particular regression target.

1 Introduction

Researchers often reuse one predicted item label across many records and construct several aggregates from those records. A website's content label can enter the browsing histories of many people; a merchant classification can enter many households' expenditures; a document label can contribute to several measures of an organization or period. The resulting totals and shares may become regression outcomes, covariates, or both. A mistake made once at the item level then appears repeatedly in the analysis.

The consequences depend on the reported quantity. An error on an item used widely by two groups can move both groups' means while largely canceling from their difference. The same error can change several regression covariates together. If the constructed measure is an outcome, a least-squares coefficient is linear in its item labels. If it is a covariate, changing a label changes the regression's numerator and denominator. Neither the number of mistakes nor the number of downstream records describes these effects by itself.

The distinction between accurate classification and accurate aggregate estimation is established. Quantification addresses class prevalence; cost quantification already allows nonnegative item weights (Forman, 2008; Hopkins and King, 2010; Vaz et al., 2019). Research on predicted covariates and outcomes shows that substituting predictions can change regression coefficients and inference (Fong

*gsood07@gmail.com. The observation that errors on popular domains can distort browsing totals appears in Sood (2018) and Laohaprapanon and Sood (2020). This paper develops its implications for aggregates and regression. Code: `https://github.com/finite-sample/total_error`.

and Tyler, 2021; Wang et al., 2020). Classification followed by aggregation into regression indices is also covered by recent generated-variable work (Battaglia et al., 2025; Christensen and Hansen, 2026). The present question is how those inferential claims change when an uncertain item supplies parts of many regression observations and several constructed variables.

Two other literatures clarify this question. Generated-regressor and measurement-error methods account for estimated calibration, correlated errors, and uncertainty in multiple estimation stages (Pagan, 1984; Murphy and Topel, 1985; Shaw et al., 2021; Boe et al., 2024). Shared-source inference explains why records with overlapping loadings need not provide independent information (Adão et al., 2019; Borusyak et al., 2022; Owen and Eckles, 2012). These results require particular centering, sampling, or shock-assignment assumptions. A common covariance structure alone does not establish those assumptions for classification errors. Conversely, a biased coefficient does not imply an inconsistent standard error: generated-covariate models include regimes in which conventional robust SEs estimate variability correctly but ignore first-order centering bias (Christensen and Hansen, 2026; Battaglia et al., 2025).

I formalize reuse by matrices mapping item labels into record-level measures. This representation yields the error in a linear estimate, the joint covariance across estimates, and the normal-equation distortion when an aggregate is a regression covariate. It also identifies the validation unit. Verifying one item can correct pieces of many records without revealing any record's complete covariate. Existing validation-sample regression methods, survey difference estimators, and prediction-powered inference provide correction principles (Shepherd et al., 2012; Lumley et al., 2011; Angelopoulos et al., 2023; Egami et al., 2023; Kluger et al., 2025). Applying them here requires preserving that item-level construction, including label pairs in quadratic regression moments.

The baseline target is the coefficient that complete labels would produce in the observed panel. For an existing probability sample of verified items, I give moment corrections and an estimable joint covariance, then state the conditions needed to propagate them into regression SEs. The corrections use standard finite-population methods, including the treatment of quadratic totals (Sitter and Wu, 2002). Their exact unbiasedness does not guarantee normal coefficient intervals. I also show what is lost when verified errors are reduced to confusion counts and distinguish these fixed-label results from population interpretations based on calibrated conditional means.

Simulations compare coefficient bias, empirical variability, estimated SEs, and interval coverage for aggregates used on either or both regression sides. A June 2022 YouGov browsing sample supplies an empirical illustration of reuse and visit concentration. Its imposed errors are controlled examples, not estimates of a fitted classifier's performance. A separate comparison of observed adult-domain classifications shows how actual label disagreement changes weighted shares without treating either classifier as verified truth.

2 Items, records, and inferential targets

There are k distinct items, with reference labels $y_j \in \{0,1\}$ and predictions $q_j \in \{0,1\}$. A known $n \times k$ matrix C_r maps them into the rth record-level measure:

$$T_r = C_r y, \qquad S_r = C_r q, \qquad d_r = S_r - T_r = C_r e, \qquad e = q - y.$$

Rows can represent people, places, organizations, or periods; columns represent the labeled items. Entries can be counts, durations, expenditures, or observed shares. An item may contribute to several rows and several matrices. Different aggregates therefore share the same vector e.

For fixed record weights a, a linear target and its prediction-based estimate are

$$\theta = a^{\mathsf{T}} C_r y = w^{\mathsf{T}} y, \quad \widehat{\theta} = w^{\mathsf{T}} q, \quad w = C_r^{\mathsf{T}} a. \tag{1}$$

Totals have nonnegative weights; comparisons can have signed weights. Proportions with observed denominators fit this expression after the denominators are absorbed into C_r. A denominator depending on unknown labels, a regression using $C_r q$ as a covariate, or another nonlinear transformation instead has error $F(C_1 q, \ldots, C_R q, Z) - F(C_1 y, \ldots, C_R y, Z)$, where Z contains other observed inputs. For linear targets the exact error is

$$\Delta = \widehat{\theta} - \theta = w^{\mathsf{T}} e. \tag{2}$$

For illustration, consider three records and two items with

$$C = \begin{pmatrix} 4 & 0 \\ 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad y = (0,1)^{\mathsf{T}}, \qquad q = (1,1)^{\mathsf{T}}.$$

One false positive changes the record measures from $T = (0,2,1)^{\mathsf{T}}$ to $S = (4,3,1)^{\mathsf{T}}$. With $Z = (0,1,2)^{\mathsf{T}}$ and an intercept, regressing T on Z gives slope $1/2$; substituting S gives $-3/2$. Regressing Z on T also gives $1/2$, while regressing Z on S gives $-9/14$. The same item mistake has different consequences in the two roles.

The primary target below is a descriptive coefficient for this observed panel with complete item labels. The matrices, observed regressors, and observed outcomes are held fixed when calculating its label error. Uncertainty requires an additional repeated experiment. A prediction-error model varies q through a specified fitting or prediction process while holding y fixed. An item-validation design instead varies which fixed labels are verified while holding q fixed. Section 4 uses the latter to construct feasible corrections. Neither experiment by itself establishes inference about a population or causal coefficient.

If records are sampled as well, their sampling distribution must be stated. An SE conditional on frozen predictions concerns that record-sampling problem. It need not quantify uncertainty about the complete-label coefficient. The corresponding variance components can be dependent; adding a label-error variance to a reported regression variance is justified only after deriving their joint distribution.

3 Regression coefficients and their uncertainty

3.1 An aggregate used as the outcome

With a fully observed, full-rank design X, let $A = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}$. Regressing an aggregate outcome Cy on X gives $\beta_y = ACy$; using Cq gives $\beta_q = ACq$. Hence

$$\beta_q - \beta_y = ACe, \qquad \mathrm{E}[\beta_q - \beta_y] = ACb, \qquad \mathrm{Cov}(\beta_q - \beta_y) = AC\Sigma C^{\mathsf{T}} A^{\mathsf{T}}, \tag{3}$$

where $b = \mathrm{E}[e]$ and $\Sigma = \mathrm{Cov}(e)$ refer to a stated prediction-error process conditional on the fixed panel. A coefficient contrast has item weights $w = C^{\mathsf{T}} A^{\mathsf{T}} r$. Equations (2) and (3) therefore cover outcome-regression coefficients as well as means and group comparisons.

Even independent errors on distinct items need not give independent errors across records. If $\Sigma = \mathrm{diag}(v_j)$,

$$\mathrm{Cov}(d_i, d_\ell) = \sum_j c_{ij} c_{\ell j} v_j. \tag{4}$$

The same item shock appears wherever its loading is nonzero. A record-level variance calculation that discards these covariances can be too small for positive averages or too large for signed contrasts.

The exact direction depends on the coefficient's item weights, not just item popularity. This is the common-source structure studied in shift-share inference; its identification and asymptotic assumptions still need to hold in the particular application (Adão et al., 2019; Borusyak et al., 2022).

Equation (3) quantifies the label-error component. For a population regression that also treats the complete-label outcome as random, write $T = X\beta + u$. With $\Omega = \mathrm{Cov}(u)$ and $K = \mathrm{Cov}(u, e)$, conditional on X, C,

$$\mathrm{Var}(\beta_q) = A\{\Omega + C\Sigma C^\mathsf{T} + KC^\mathsf{T} + CK^\mathsf{T}\}A^\mathsf{T}. \tag{5}$$

The cross terms vanish under additional restrictions such as independence of u and e. Estimating only $C\Sigma C^\mathsf{T}$ does not estimate this whole variance. Corrections for predicted outcomes must specify the relationship between the observed outcome, its prediction, and the downstream regressors (Wang et al., 2020).

3.2 An aggregate used as a covariate

Let Y be an observed outcome and let V be fully observed controls, including an intercept. Put $M_V = I - V(V^\mathsf{T}V)^{-1}V^\mathsf{T}$. The complete-label coefficient on $T = Cy$ is

$$\beta(y) = \frac{y^\mathsf{T}C^\mathsf{T}M_V Y}{y^\mathsf{T}C^\mathsf{T}M_V Cy} = \frac{N(y)}{D(y)}, \qquad D(y) > 0. \tag{6}$$

For $t = M_V Cy$, $h = M_V Ce$, and $u = M_V Y - \beta(y)t$, the exact change is

$$\beta(q) - \beta(y) = \frac{h^\mathsf{T}u - \beta(y)\{t^\mathsf{T}h + h^\mathsf{T}h\}}{(t+h)^\mathsf{T}(t+h)}, \tag{7}$$

provided the predicted covariate has positive residual variation. The denominator changes as well as the numerator, and the linear and quadratic error terms need not have the same sign. Ordinary attenuation is therefore not a general description.

For small perturbations, the coefficient's item gradient is

$$\lambda = \frac{C^\mathsf{T}M_V Y - 2\beta(y)C^\mathsf{T}M_V Cy}{D(y)}. \tag{8}$$

Thus $\lambda^\mathsf{T}b$ and $\lambda^\mathsf{T}\Sigma\lambda$ are first-order bias and variance expressions only when the remainder in equation (7) is negligible on the relevant scale. They are not feasible SEs unless the unknown labels and error moments can be replaced by sufficiently accurate estimates. Weak residual variation in Cy, or a few influential items, can make the linear approximation poor.

These are generated-covariate problems in the sense of Fong and Tyler (2021), Battaglia et al. (2025), and Christensen and Hansen (2026). Reuse makes their input construction and validation unit explicit. A verified item changes pieces of several rows; it usually does not reveal any row's complete covariate. Results based on validated complete rows require an adaptation or a different validation design (Kluger et al., 2025). An item with the same loading in every row has $M_V C_{\cdot j} = 0$, so its error cancels from this coefficient even if it dominates unadjusted totals.

3.3 Several constructed variables

If both regression sides use the same predictions, the numerator in equation (6) becomes $y^\mathsf{T}C_1^\mathsf{T}M_V C_0 y$. Shared errors now affect both regressor–outcome and regressor–regressor cross-products. For constructed covariates $C_r y$, the repeated-prediction error cross-product is

$$\mathrm{E}[d_r^\mathsf{T}M_V d_s] = b^\mathsf{T}C_r^\mathsf{T}M_V C_s b + \mathrm{tr}(C_r^\mathsf{T}M_V C_s \Sigma). \tag{9}$$

It can be nonzero with independent item errors. Correcting the marginal error of each constructed variable does not determine these joint terms. Correlated outcome and exposure errors are already studied in measurement error models (Shaw et al., 2021); here their relationship follows from the shared item labels and the two loading matrices.

3.4 Conditions for standard-error interpretations

Bias, variability, and coverage must be assessed separately. Some generated-covariate asymptotic regimes have consistent conventional robust SEs but a biased limiting distribution around the desired coefficient (Christensen and Hansen, 2026; Battaglia et al., 2025). Reuse does not rule out such a regime. It requires checking its laws of large numbers, central limit conditions, and variance-estimation assumptions for the actual loadings.

For a linear estimate with independent item errors, put $s^2 = \sum_j w_j^2 v_j$. The sufficient Lindeberg condition $\max_j |w_j|/s \to 0$ gives a normal limit for centered label error. It need not hold as the number of records grows: those records may continue to load on the same few uncertain items. Even a small maximum variance share is insufficient when errors become rare (Appendix J). Clustering by record does not recover cross-record item covariance, while assigning each record one item cluster omits its other memberships. Resampling must preserve the sources that the inferential model treats as random (Owen and Eckles, 2012). Shared classifier fitting can additionally correlate errors on distinct items; that is a separate mechanism from reusing one item's error.

4 Validation information and feasible inference

4.1 An existing probability sample of items

Suppose predictions are frozen before items are selected for verification. The audit observes y_j for selected items, with known inclusion probabilities. This is one source of inference about the complete-label panel coefficient. It does not require optimal selection or calibrated scores, but it does require correctly verified labels and a known design. Audit-based regression correction predates current surrogate-assisted methods (Shepherd et al., 2012); survey calibration and design-based supervised learning provide related estimating-function constructions (Lumley et al., 2011; Egami et al., 2023).

For a linear target, let R_j indicate verification and let $\pi_j = \Pr(R_j = 1) > 0$. The difference estimator is

$$\widetilde{\theta} = w^\mathsf{T} q + \sum_j \frac{R_j}{\pi_j} w_j (y_j - q_j). \tag{10}$$

It is design unbiased for $w^\mathsf{T} y$. For several targets, replace w_j by its vector of weights across targets. This preserves the joint correction contributed by one verified item. The exact design covariance uses the first- and second-order inclusion probabilities, not a separate error draw for each downstream record (Appendix B).

4.2 Regression moments and their joint covariance

For a constructed covariate, equation (6) has a linear numerator and quadratic denominator. If both regression sides are constructed, both moments are quadratic. More generally, each component of the normal equations has form $m_s(y) = a_s^\mathsf{T} y + y^\mathsf{T} G_s y$, with G_s symmetric. Writing $r = y - q$ gives

$$\boldsymbol{m}(y) = \boldsymbol{m}(q) + \sum_{A \in \mathcal{A}} c_A, \qquad \widehat{\boldsymbol{m}} = \boldsymbol{m}(q) + \sum_{A \in \mathcal{A}} \frac{R_A}{p_A} c_A, \tag{11}$$

where $\mathcal{A}$ contains singleton items and unordered item pairs, $R_A = \prod_{j \in A} R_j$, and $p_A = \Pr(R_A = 1) > 0$. The sth singleton contribution is $r_j\{a_{sj} + 2(G_s q)_j\} + G_{s,jj} r_j^2$; the pair contribution for $j < \ell$ is $2G_{s,j\ell} r_j r_\ell$. Both are observed whenever their items are verified. Quadratic finite-population totals can be treated as totals on such a derived population (Sitter and Wu, 2002).

Proposition 1 (Joint audit moments)**.** *Conditional on the panel and frozen predictions,* $\mathrm{E}_R[\widehat{\boldsymbol{m}}] = \boldsymbol{m}(y)$*. If every relevant* $p_{A\cup B}$ *is known and positive, an unbiased estimator of its covariance is*

$$\widehat{V} = \sum_{A,B \in \mathcal{A}} \frac{R_{A\cup B}(p_{A\cup B} - p_A p_B)}{p_A p_B p_{A\cup B}} c_A c_B^{\mathsf{T}}. \tag{12}$$

Thus quadratic regression moments require inclusion probabilities for sets of up to four items.

Proof. Each term in equation (11) has expectation c_A. The covariance of the A and B terms is $(p_{A\cup B} - p_A p_B) c_A c_B^{\mathsf{T}} / (p_A p_B)$. Multiplication by $R_{A\cup B}/p_{A\cup B}$ gives an unbiased estimate of each term. □

This is a Horvitz–Thompson covariance calculation, applied jointly to the regression moments. It does not introduce a new regression estimator. Independent Bernoulli item sampling gives a particularly simple implementation because disjoint item sets have zero covariance; Appendix F gives the expression used below.

For $\beta = N/D$, the gradient of the regression map is $g = (1/D, -N/D^2)^{\mathsf{T}}$. A feasible delta SE uses

$$\widehat{\beta} = \widehat{N}/\widehat{D}, \qquad \widehat{\mathrm{se}}(\widehat{\beta}) = \{\widehat{g}^{\mathsf{T}} \widehat{V} \widehat{g}\}^{1/2}. \tag{13}$$

A normal interval is justified when the joint moments have a normal approximation, the denominator stays sufficiently far from zero, the linearization remainder is negligible, and the estimated variance is consistent on the coefficient's variance scale. With several constructed covariates, the moment vector includes all regressor–outcome and regressor–regressor products; the derivative of the full regression map must be applied to their joint covariance.

These conditions do not follow from unbiasedness of either moments or covariance. In a finite audit, $\widehat{D}$ can be nonpositive and an unbiased covariance estimate can be indefinite. Such draws do not support the stated Wald SE. A few influential items can also leave a nonnormal coefficient distribution even when the covariance formula is exact. Section 5 evaluates both SE estimation and coverage, including these failures.

A finite-sample alternative starts with a confidence set $\mathcal{Y}_A$ for the complete label vector, such as the simultaneous count certificate in Appendix B. On its coverage event,

$$\beta(y) \in \{\beta(z) : z \in \mathcal{Y}_A,\ D(z) > 0\}.$$

The coverage implication holds for any fixed downstream map; it is the usual projection of a confidence set, also used in prediction-powered inference (Angelopoulos et al., 2023). Computing useful nonlinear endpoints and handling possible rank failures are additional tasks. The resulting set may be very wide or unbounded.

4.3 Calibrated predictions and estimated stages

Calibration is not required for fixed-label bounds or the probability audit above. A score-based population interpretation needs different assumptions. If information $\mathcal{H}$ includes the loading matrices and controls and $p_j = \mathrm{E}[y_j \mid \mathcal{H}]$, then $Cp = \mathrm{E}[Cy \mid \mathcal{H}]$. Under $Y = V\gamma + \beta C y + \varepsilon$,

with $\mathrm{E}[\varepsilon \mid \mathcal{H}] = 0$ and a full-rank conditional-mean design, regressing Y on (V, Cp) identifies the population β. Ordinary calibration across items, $\Pr(y = 1 \mid p = s) = s$, does not establish these conditional means or the exclusion restriction. Appendix C gives a weighted counterexample and separates expected aggregates from realized finite-item targets.

Even correct conditional means do not supply the covariance of the remaining errors across records. Estimating calibration adds another estimation stage. A stacked estimating-equation covariance or a justified bootstrap must propagate that stage together with the downstream fit (Murphy and Topel, 1985; Boe et al., 2023, 2024). Methods that validate complete regression rows and permit nonuniform sampling offer useful precedents (Kluger et al., 2025); the sampled unit must still match the verified information available here.

If uncertainty instead concerns repeated classifier fitting, the fitting sample and all item predictions must be varied jointly under a resampling scheme justified for that process. Independently perturbing each downstream record invents independent versions of a shared label. No known sampling probabilities, error model, or training experiment can be recovered from a convenience confusion table alone.

4.4 What counts omit

Exact target-item false-positive and false-negative counts constrain which labels can be wrong, but discard their locations in the loading matrices. For linear targets, Proposition 2 gives the sharp interval obtained by sorting their item weights. Counts suffice within a predicted class only when all relevant weights are equal, or its error count is zero or exhaustive (Proposition 3). These bounds concern identification from a summary, not a standard error. Their proof and the audit count certificate are in Appendices A and B.

5 Simulation: bias, standard errors, and coverage

The simulation asks whether an uncertainty calculation for the chosen complete-label coefficient matches its repeated error. Each panel fixes the item labels, loading matrices, controls, and any observed outcome. Across repetitions, item predictions are generated by independent flips with probability 0.15, and each item is independently verified with probability 0.70. The audit probabilities and predictions are known when applying equations (11)–(13). This experiment isolates label prediction and verification; it does not sample a new population of records in each repetition.

There are 200 or 800 records. The diffuse design uses 200 items; record-specific multinomial counts depend on an observed characteristic and item features. The concentrated design gives one item three times the baseline popularity of all other items combined. A third design uses only 30 items. Counts are converted to visit shares; a second aggregation reweights them by item-specific durations. We examine an aggregate outcome, an aggregate covariate with an observed outcome, and aggregates on both regression sides. Appendix E gives the construction. The complete-label coefficient is calculated separately for every fixed panel and regression role.

The record HC method attaches HC0 SEs to the uncorrected proxy regression. Those intervals concern the proxy fit under its usual sampling interpretation; here we evaluate their use for the complete-label target. The item-audit method corrects the regression moments and estimates their joint covariance. An oracle version uses the same corrected estimate and the true conditional moment covariance, with the regression derivative evaluated at the true moments. It isolates variance estimation from linearization and distributional failures and is not a feasible comparator for an application without complete labels.

Table 1: Regression inference with reused labels in simulated panels of 800 records

Role	Loadings	Method	Bias	SD	Mean SE	Cover (%)
Outcome	Diffuse	Record HC	-0.101	0.020	0.003	0.0
Outcome	Diffuse	Item audit	0.000	0.015	0.014	93.9
Outcome	Concentrated	Record HC	-0.076	0.103	0.004	0.2
Outcome	Concentrated	Item audit	0.002	0.070	0.030	89.9
Covariate	Diffuse	Record HC	0.351	0.090	0.014	0.0
Covariate	Diffuse	Item audit	0.004	0.040	0.038	93.2
Covariate	Concentrated	Record HC	-0.379	1.288	0.020	0.0
Covariate	Concentrated	Item audit	-0.097	0.524	0.045	89.9
Both	Diffuse	Record HC	-0.001	0.048	0.008	26.4
Both	Diffuse	Item audit	0.002	0.026	0.023	92.2
Both	Concentrated	Record HC	0.009	0.035	0.008	36.2
Both	Concentrated	Item audit	0.000	0.018	0.016	90.9

Each row summarizes 1,000 repetitions of predictions and independent item verification in one fixed panel. Outcome, covariate, and both denote the roles of label-derived measures. There are 200 items; the concentrated design gives one item a large common loading. Bias and empirical SD are relative to the complete-label panel coefficient. Mean SE averages the reported HC0 or item-audit delta SE. Coverage uses a normal 1.96 cutoff. The largest binomial Monte Carlo SE across the simulation is 1.6 percentage points. No coefficient or SE is unavailable in the displayed designs.

In the diffuse 800-record panels, item-audit coverage is 93.9% for an aggregate outcome, 93.2% for an aggregate covariate, and 92.2% when both sides are constructed (Table 1). In the covariate case, the uncorrected coefficient's bias is 0.351, compared with 0.004 after moment correction. Bias and variance estimation thus answer distinct questions even in this favorable loading pattern.

Concentration makes the delta approximation less reliable. In the concentrated covariate example, the corrected coefficient has empirical SD 0.524, while its mean estimated SE is 0.045 and coverage is 89.9%. Occasional small estimated denominators produce large coefficient errors. Even the oracle delta interval covers only 91.7% in this design. With 30 items and both sides constructed, feasible coverage is 76.8%. These are results for the stated panels and error process. They show why an exact, unbiased moment covariance is insufficient to guarantee normal coefficient intervals. The smaller panels and oracle comparisons appear in Appendix E.

6 Applications

6.1 Reuse in a June 2022 browsing sample

The YouGov Pulse/RealityMine browsing sample contains 1,132 users, 34,078 domains, 103,853 user–domain records, and 4,767,099 visits. Domain labels recur across users, and visits are concentrated: the most widely used domain appears in 1,080 users' records. One domain accounts for 20.7% of visits, while the most visited 1% of domains account for 83.4%. These are properties of this observed sample, before a prediction-error model is introduced.

We use seven recorded Blacklight scanner outcomes as reference labels for an illustration. Two count outcomes are dichotomized at a positive value; the other five are binary. Shares use visits to domains with recorded scanner labels as their denominator. The scanner labels measure recorded tracking behavior rather than every form of tracking. Regressions include an intercept and indicators for gender, race, education, and age groups; the omitted categories are Male, White, HS or Below, and under 25. Their twelve non-intercept coefficients are descriptive for this panel.

To illustrate information loss from counts, we impose errors by randomly flipping the integer part

of 1%, 5%, or 10% of the domain labels within each reference class. At the 1% setting, exact false-positive and false-negative counts leave the signs of all 84 visit-share coefficients unidentified. The bound is computed separately for each coefficient, so the extreme allocations need not hold jointly. For the female coefficient on Facebook-pixel visit share, the reference value is 0.828 percentage points and the random predictions differ by 0.040 percentage points. Yet their count-compatible range for the reference coefficient is $[-17.00, 16.72]$ percentage points. Appendix D reports the outcome-specific bounds. This calculation describes what the imposed error counts fail to exclude; it does not estimate a trained classifier's likely error.

The same observed visit matrix illustrates a constructed covariate. For each other scanner outcome, we regress user-level visit share on advertising-tracker visit share and the demographic controls. We hold the outcome labels fixed and replace only the advertising-tracker labels with the 1% random-error draw. The third-party-cookie slope changes from 0.682 to 0.622; the six slopes can rise or fall (Table 2). These are coefficient changes under one imposed draw. The coverage experiment in Section 5 addresses the separate SE question.

Table 2: An aggregate covariate under imposed label errors in the browsing sample

Outcome	Recorded labels	Imposed errors
Third-party cookies	0.682	0.622
Canvas fingerprinting	0.089	0.060
Session recording	0.048	0.052
Key logging	0.063	0.063
Facebook pixel	0.140	0.113
Google Analytics	0.016	-0.010

Each row regresses the listed scanner outcome's user-level visit share on advertising-tracker visit share and demographic indicators in the June 2022 YouGov sample. Recorded labels uses the scanner's advertising-tracker labels. Imposed errors flips the integer part of 1% of those domain labels within each reference class in one random draw. Outcome labels and controls are fixed. The table describes coefficient sensitivity, not repeated-sample uncertainty or fitted-classifier performance.

The matrix also shows how reuse affects the label-error covariance under a stipulated model. Suppose distinct domain errors are independent with equal variance. Equation (J9) compares a statistic's label-error variance with a calculation that keeps each user's marginal variance but discards covariance between users. The ratio for mean visit share is 323.4; one domain contributes 81.2% of the item-level variance. The female and Black coefficient ratios are 2.12 and 0.77, respectively. The latter illustrates cancellation in a signed contrast.

Under this same model, errors in the female and male mean visit shares have correlation 0.987. Their difference has only 1.36% of the error variance obtained by treating the two group-mean errors as independent. These calculations use the observed loadings and an assumed error covariance. They diagnose the consequence of dropping reuse; they do not estimate an actual classifier's SE. A regression correction for actual predictions would additionally require the validation information described in Section 4.

6.2 Observed disagreement between domain classifiers

A separate comparison uses observed outputs from the adult-domain study of Sood and Shen (2024). Its public domain summary records 6,236,834 visits to 64,074 domains in the June 2022 YouGov panel. The study used a Piedomains rule that marks a domain when its predicted class is "porn" and its reported class probability is at least 0.033. We compare that rule with recorded "porn" categories from Bitdefender, "sex" from Forcepoint, and "Pornography" from alphaMountain. Each

row retains only domains with a recorded category from that provider, and visit shares use the visits in that row's frame as their denominator. Missing provider categories are not negative labels.

Table 3: Observed classifier disagreement and visit-weighted shares

Reference	Domains	Visits (%)	Agree (%)	Domain + (%)	Visit + (%)	Count range (%)
Bitdefender	43,391	84.2	97.7	1.92/3.36	1.05/1.71	[0.07, 82.11]
Forcepoint	47,950	87.5	98.0	1.65/3.04	1.02/1.64	[0.07, 80.61]
alphaMountain	36,063	52.7	99.0	0.98/1.35	1.38/2.16	[0.04, 68.07]

Piedomains is the proxy in every row; the named provider supplies the reference category. "Visits" is the share of all 6,236,834 visits represented by domains with a recorded reference category. Agreement counts domains equally. "Domain +" and "Visit +" list Piedomains/provider percentages, first for domains counted equally and then for visits within that row's frame. The count range is the sharp range of the provider visit share if only the two directional disagreement counts are kept, holding Piedomains labels and visit weights fixed. The providers use different category definitions; none is an adjudicated truth label. A recorded category other than the named positive one counts as negative for this comparison, including nonspecific categories such as "misc."

On the 43,391 domains categorized by Bitdefender, which account for 84.2% of visits, the two classifications agree on 97.7% of domains. Piedomains marks 1.92% of domains positive, compared with 3.36% under Bitdefender's category. Their visit-weighted shares are 1.05% and 1.71%, respectively. The discrepancy is visible before weighting domains by visits, so this comparison does not isolate an effect of visit concentration. There are 193 Piedomains-only and 816 Bitdefender-only positive domains.

With both classifications observed, the provider's visit share is known. If the disagreement locations were discarded and only their directional counts retained, its compatible range would be [0.07, 82.11]%. A single Piedomains-negative domain carries 18.8% of visits in this frame and can dominate the count-only upper bound. Appendix L shows how that range changes when the observed label pairs of the most-visited domains are retained. This isolates information discarded by a confusion table; it does not establish that the extreme allocations are plausible classifier errors.

The comparison measures disagreement between classification systems whose categories differ. It does not estimate either system's error against verified content labels. The domain-level comparison does not provide regression SEs for individual browsing outcomes. Applying the item-audit correction would require compatible person–domain links, verified labels under a declared content rule, and the label-selection probabilities. The Piedomains score and provider categories alone do not establish the conditional probabilities required in Section 4.3.

7 Discussion

Reusing predictions changes the link between classification error and a reported estimate. The relevant weights depend on the comparison, controls, and regression role. Reuse can link errors across records and across constructed variables even when distinct item errors are independent. An outcome coefficient is linear in those errors, while a constructed covariate changes the regression's cross-products and denominator.

This construction connects familiar inference problems. Generated-variable methods address coefficient centering and estimated stages; shared-source methods address dependence and influential loadings; validation and survey methods address missing verified information. Their assumptions must be checked for the same item-to-record map. A larger regression sample does not itself supply more independently labeled or verified items.

For a probability audit, item and item-pair contributions give unbiased regression moments and

an unbiased joint covariance estimate. The normal coefficient interval is a further approximation. The simulations show substantial reductions in coefficient bias but also undercoverage in concentrated and few-item designs. An exact covariance formula cannot remove a weak denominator or create a central limit theorem. Reporting these distinctions makes clear which correction has been justified and which interval interpretation remains model dependent.

Confusion counts and ordinary score calibration preserve different information. Counts generally omit the error locations needed by a weighted estimate. Calibrated conditional means can support population plug-ins under the relevant regression assumptions, but calibration averaged over items does not establish those assumptions or the joint error covariance. The validation design and retained item identities therefore determine which downstream adjustments and uncertainty statements the data support.

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Appendix

A Sharp bounds from confusion counts

A confusion matrix can be exactly correct for the target items and still fail to determine a downstream estimate. This is an information problem before it is a sampling problem. Fix the predictions q and the item weights w in equation (1). We now ask what a validation summary establishes about the fixed target $\theta = w^\top y$, without specifying a stochastic model for how the classifier's mistakes were generated.

Let $P = \{j : q_j = 1\}$ and $N = \{j : q_j = 0\}$. Suppose the entire target-population confusion matrix is known: f predictions in P are false positives and g predictions in N are false negatives. This grants more information than ordinary held-out validation, whose error rates require a transport assumption. For a finite collection B of weights, let $S_-(B, m)$ and $S_+(B, m)$ denote the sums of its m smallest and largest elements, with a zero sum when $m = 0$.

Proposition 2 (Sharp bounds from confusion counts)**.** *Across binary label vectors compatible with q, f, g, the smallest and largest possible errors $\Delta = \widehat{\theta} - \theta$ are*

$$\begin{aligned} L &= S_-(w_P, f) - S_+(w_N, g), \\ U &= S_+(w_P, f) - S_-(w_N, g). \end{aligned} \tag{A1}$$

Consequently $[\widehat{\theta} - U, \widehat{\theta} - L]$ is the smallest interval containing every compatible value of θ. Both endpoints are attainable.

Proof. Only false positives and false negatives contribute: $\Delta = \sum_{j \in \mathrm{FP}} w_j - \sum_{j \in \mathrm{FN}} w_j$. Select f indices from P and g from N. The two selections have disjoint supports and no other restrictions. Selecting the smallest positive-stratum weights and largest negative-stratum weights minimizes the error; reversing these selections maximizes it. Assign the selected indices the corresponding true labels to attain each endpoint. □

The compatible values form a finite set, so the interval is its convex hull; not every interior value need be attainable. A reported sign is established by these counts only if the bounding interval excludes zero. The calculation requires sorting, rather than a general integer-programming solver.

For example, take $q = (1, 1, 0, 0)$, $w = (1, -1, 1, -1)$, and $f = g = 1$. The same confusion matrix permits errors of -2, 0, and 2. Prevalence, accuracy, FPR, and FNR are identical across those assignments. Adding correctly classified items with zero weight makes accuracy arbitrarily close to one without resolving the target. In the browsing application, these can be domains that affect individual exposure but cancel from a group contrast.

Proposition 3 (When counts suffice)**.** *In a stratum of M items sharing the same predicted label, with exactly m errors, the error sum is point identified if and only if $m \in \{0, M\}$ or all its contrast weights are equal. Its identification width obeys*

$$S_+(w, m) - S_-(w, m) \le \min(m, M - m)\{\max_j w_j - \min_j w_j\}. \tag{A2}$$

Widths add across strata. For several comparisons, counts determine every comparison if and only if their vector of item weights is constant within each stratum with an interior error count.

Proof. Constant weights, no errors, and all errors each determine the sum. If $0 < m < M$ and two weights differ, choose an error set containing one of those indices but not the other. Exchanging

them changes the sum, proving necessity. The difference between two sums of m terms is at most m times the range. Taking complements gives the corresponding bound with $M - m$ terms. The vector statement applies the exchange argument to each coordinate. □

For a stratum mixing predicted labels, the corresponding criterion applies to the signed contributions $(2q_j - 1)w_j$, not to w_j alone. The result specifies what a finer validation table must preserve. Stratifying by predicted label alone is sufficient only when the contrast attaches equal weights within each partially erroneous stratum. Partitioning items by similar contrast weights can reduce ambiguity; bins chosen for one comparison need not work for another. In browsing, error rates conditional on user demographics are not automatically available: the same domain can enter many groups.

B Audit count bounds and linear corrections

An existing probability sample of verified target items can recover information that a confusion matrix discards; it need not have been allocated optimally. The construction uses established finite-population counting and audit ideas (Stark, 2008; Shekhar et al., 2023). Partition the target items using information known before labels are audited, then draw a fixed-size random sample without replacement within each stratum. Exact hypergeometric inversion gives a confidence interval $[\ell_h, u_h]$ for the number of incorrect predictions in stratum h; allocate error probabilities across strata so their sum is at most δ.

Let O contain the audited items and $\mathcal{Y}_A$ the binary label vectors that match their verified labels and whose error counts in each stratum lie in $[\ell_h, u_h]$. Its construction and the sorting formula for its extremes are in Appendix K.

Proposition 4 (A simultaneous count certificate)**.** *For any fixed item weights w, the interval $[\min_{z\in\mathcal{Y}_A} w^\mathsf{T} z, \max_{z\in\mathcal{Y}_A} w^\mathsf{T} z]$ covers the fixed target $w^\mathsf{T} y$ with probability at least $1-\delta$ over the audit design. The same event covers every linear contrast of this one label vector. Its endpoints are sharp given the audited labels and the count intervals.*

The certificate is simultaneous because the count event constrains the same label vector for every w. It can nevertheless be wide: it retains error counts but discards the weights attached to unaudited error locations. The example and full audit comparison in Appendix K show this loss. Those simulations evaluate a design chosen for one coefficient; they do not establish efficient simultaneous auditing for other contrasts.

For one declared contrast, a stratified difference estimator instead keeps the audited item identities and their weighted residuals:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j\in O_h} w_j(y_j - q_j). \tag{B1}$$

With fixed predictions and verified labels, this estimator is design unbiased. A normal interval needs a separate approximation. The conservative finite-sample Hoeffding interval in the appendix is wide in this application; the paper does not establish an efficient finite-sample interval or an optimal audit allocation. The distinction here is between preserving the locations of audited errors and reducing them to counts.

The same collected labels can correct several estimates jointly. Let W contain their item-weight vectors, let R_j indicate that item j was verified, and let $\pi_j > 0$ be its known inclusion probability.

Set $g_j = (y_j - q_j)W_{j\cdot}^{\mathsf{T}}$. With predictions frozen before the validation sample was selected,

$$\widetilde{\boldsymbol{\theta}} = W^{\mathsf{T}}q + \sum_j \frac{R_j}{\pi_j} g_j, \qquad \mathrm{E}_R[\widetilde{\boldsymbol{\theta}}] = W^{\mathsf{T}}y. \tag{B2}$$

The reusable item, not each downstream record, is the sampling unit. If $\pi_{j\ell} = \Pr_R(R_j = R_\ell = 1)$ and $\pi_{jj} = \pi_j$, the exact joint design covariance is

$$\mathrm{Cov}_R(\widetilde{\boldsymbol{\theta}}) = \sum_{j,\ell} \frac{\pi_{j\ell} - \pi_j\pi_\ell}{\pi_j\pi_\ell} g_j g_\ell^{\mathsf{T}}. \tag{B3}$$

If every relevant $\pi_{j\ell} > 0$, replacing each summand by $R_jR_\ell/\pi_{j\ell}$ times that summand gives an unbiased covariance estimator. For independent Bernoulli item selection, only the diagonal terms remain, giving $\sum_j (1-\pi_j)(y_j - q_j)^2 W_{j\cdot}^{\mathsf{T}} W_{j\cdot}/\pi_j$. The off-diagonal entries of this covariance describe how a single verified item corrects several reported estimates together. Highly influential items with small π_j can leave a valid estimator with a very unstable variance estimate and a poor normal approximation.

If validation items were chosen purposively and their inclusion probabilities are unknown, these design claims do not apply. One can instead state a model for how validation errors transfer to the target items, or report bounds under explicit restrictions. A confusion matrix from a convenience set does not identify either the target-item correction or its standard error. The ordinary regression standard error conditional on frozen predictions answers a different question; it does not include the error from unverified labels. If the same selected labels were used to fit the predictions, the frozen-proxy design argument also needs modification; cross-fitting or joint analysis of training and validation would require its own justification.

C Score calibration and conditional means

The bounds from exact target-item confusion counts and the probability audit above condition on fixed predictions and labels. Neither requires calibrated classifier scores. The audit needs a known probability-sampling design for target items and verified labels for the sampled items. If the error counts come instead from a separate validation collection, applying them to the target items requires a transport assumption, whether or not that collection's scores are calibrated.

A score-based plug-in uses a different source of information. Let $p_j \in [0,1]$ be a score intended to represent the probability that target item j has label one, and put $\theta_p = w^{\mathsf{T}}p$. To interpret this as a conditional mean, introduce a model in which the labels are random given information $\mathcal{I}$ containing the scores and the target weights. Write $\mu_j = \mathrm{E}[y_j \mid \mathcal{I}]$. Then

$$\mathrm{E}[\theta_p - \theta \mid \mathcal{I}] = w^{\mathsf{T}}(p - \mu). \tag{C1}$$

The plug-in equals $\mathrm{E}[\theta \mid \mathcal{I}]$ if and only if this weighted error is zero. A sufficient condition is $p_j = \Pr(y_j = 1 \mid \mathcal{I})$ for each target item. For a particular target, offsetting score errors can also make the weighted sum zero without item-level conditional calibration.

Ordinary calibration, $\Pr(y = 1 \mid p = s) = s$ across items, is weaker. For example, let half the items have label one and weight one, and the other half have label zero and weight zero. Give every item score $1/2$. The scores are empirically calibrated in the unweighted item collection, but θ_p is only half of the realized target. Conditioning on features that distinguish the two weight groups, or directly checking the weighted residual, exposes the failure. Calibration assessed on a separate collection also has to transfer to the target items and their label definition.

For a browsing visit share, $w_j = v_j / \sum_\ell v_\ell$ gives $\theta_p = \sum_j v_j p_j / \sum_j v_j$. Conditional probabilities for the specified content label would make this the model's expected visit share. A group contrast or regression coefficient has different, possibly signed weights; its required weighted score condition changes accordingly.

Even valid target-item conditional probabilities identify a conditional expectation, not the realized finite-item target. Under the conditional-label model, uncertainty around that expectation depends on the joint distribution:

$$\mathrm{Var}(\theta \mid \mathcal{I}) = w^{\mathsf{T}} \mathrm{Cov}(y \mid \mathcal{I}) w. \tag{C2}$$

Item-level calibration supplies neither the off-diagonal covariances nor, by itself, a coverage guarantee for an interval about the realized target. Estimating the scores introduces further uncertainty. These model-based claims are distinct from the fixed-label, design-based coverage of the audit.

D Confusion-count bounds in the browsing illustration

Table D1: What exact confusion counts establish at 1% imposed error in each true class

Scanner outcome	Sign identified	Median realized error (pp)	Median bound width (pp)
Advertising trackers	0/12	0.43	47.8
Third-party cookies	0/12	0.09	52.5
Canvas fingerprinting	0/12	0.34	29.5
Session recording	0/12	0.40	31.6
Key logging	0/12	2.10	21.3
Facebook pixel	0/12	0.10	43.6
Google Analytics	0/12	0.33	21.6

Each row covers twelve demographic coefficients of visit shares in the browsing data. "Sign identified" counts bounding intervals excluding zero for imposed random predictions. Realized error is the median absolute difference between the prediction-based and reference coefficients. Bound width is the median identification-interval width; both are in percentage points (pp). Error counts are rounded down within each true class. These are not sampling intervals.

E Simulation construction and additional results

For each panel, the observed characteristic x_i is equally spaced on $[-1, 1]$, controls are an intercept and $\sin(i)$, and item features z_j are equally spaced on $[-1.5, 1.5]$. Labels are $y_j = 1\{z_j > 0\}$. Each record receives 80 multinomial visits with probabilities proportional to $p_j \exp(x_i z_j)$. Baseline popularity is $p_j = 1$. In the concentrated design, $p_1 = 3(k-1)$, $z_1 = 1.5$, and $y_1 = 1$. The resulting visit shares form C_1. Duration shares C_0 reweight its columns by $\exp\{\cos(j)\}$ and renormalize each row. The observed outcome in the covariate case is $Y_i = 0.8(C_1 y)_i + 0.1 x_i + 0.02\cos(i)$. These numerical constants define the simulation rather than estimate a behavioral model.

The outcome regression uses $C_1 q$ as outcome and x as covariate. The covariate regression uses the observed Y as outcome and $C_1 q$ as covariate. The both-sides regression uses $C_0 q$ as outcome and $C_1 q$ as covariate. All include the stated controls. Independent item prediction flips have probability 0.15, while independent audit inclusion has probability 0.70. Their common item indicators are reused wherever a label appears. Audit corrections condition on each realized prediction vector; their moment-unbiasedness therefore also holds when averaging over these prediction draws.

Table E1: The same simulation in panels of 200 records

Role	Loadings	Method	Bias	SD	Mean SE	Cover (%)
Outcome	Diffuse	Record HC	-0.107	0.021	0.006	0.0
Outcome	Diffuse	Item audit	0.001	0.015	0.015	94.7
Outcome	Concentrated	Record HC	-0.075	0.102	0.008	2.6
Outcome	Concentrated	Item audit	0.001	0.072	0.029	92.2
Outcome	Few items	Record HC	-0.099	0.051	0.006	2.2
Outcome	Few items	Item audit	-0.001	0.037	0.033	85.9
Covariate	Diffuse	Record HC	0.364	0.097	0.027	0.0
Covariate	Diffuse	Item audit	0.001	0.041	0.039	94.4
Covariate	Concentrated	Record HC	-0.326	1.221	0.037	0.3
Covariate	Concentrated	Item audit	-0.096	0.515	0.044	91.9
Covariate	Few items	Record HC	0.354	0.221	0.029	1.3
Covariate	Few items	Item audit	0.022	0.103	0.072	84.2
Both	Diffuse	Record HC	0.002	0.050	0.015	44.9
Both	Diffuse	Item audit	0.000	0.026	0.024	92.2
Both	Concentrated	Record HC	0.007	0.033	0.014	62.6
Both	Concentrated	Item audit	0.001	0.019	0.016	89.5
Both	Few items	Record HC	0.008	0.118	0.017	22.4
Both	Few items	Item audit	0.001	0.058	0.041	80.0

Methods, repetitions, target, and columns match Table 1. Diffuse and concentrated designs have 200 items; few-item designs have 30. All displayed coefficient and SE calculations are available. Changing the number of records does not change the number of independently predicted or verified items.

Table E2: Oracle audit delta intervals in panels of 800 records

Role	Loadings	Method	Bias	SD	Mean SE	Cover (%)
Outcome	Diffuse	Oracle audit	0.000	0.015	0.014	95.1
Outcome	Concentrated	Oracle audit	0.002	0.070	0.034	95.7
Outcome	Few items	Oracle audit	-0.001	0.037	0.035	96.9
Covariate	Diffuse	Oracle audit	0.004	0.040	0.039	95.1
Covariate	Concentrated	Oracle audit	-0.097	0.524	0.043	91.7
Covariate	Few items	Oracle audit	0.023	0.110	0.085	92.4
Both	Diffuse	Oracle audit	0.002	0.026	0.024	94.6
Both	Concentrated	Oracle audit	0.000	0.018	0.016	93.6
Both	Few items	Oracle audit	-0.003	0.062	0.049	92.8

The point estimate is the item-audit estimate. The SE uses its true conditional moment covariance and the derivative at the true moments. This comparator is infeasible without complete labels. Coverage can still depart from 95% because the coefficient is nonlinear and the corrected moment distribution need not be normal. Other definitions match Table 1.

F Computing the joint moment covariance

For independent Bernoulli verification, let L_j be the vector of singleton contributions in equation (11), and let $H_{j\ell} = H_{\ell j}$ be its pair-contribution vector, with $H_{jj} = 0$. Put $\xi_j = R_j/\pi_j - 1$ and $t_j = (1 - \pi_j)/\pi_j$. Expanding the audit error gives the orthogonal decomposition

$$\widehat{\boldsymbol{m}} - \boldsymbol{m} = \sum_j \xi_j \nu_j + \sum_{j<\ell} \xi_j \xi_\ell H_{j\ell}, \qquad \nu_j = L_j + \sum_{\ell \neq j} H_{j\ell}.$$

Independence and centering make distinct terms uncorrelated, so the oracle conditional covariance is

$$V = \sum_j t_j \nu_j \nu_j^\mathsf{T} + \sum_{j<\ell} t_j t_\ell H_{j\ell} H_{j\ell}^\mathsf{T}.$$

For a feasible calculation, define $\widehat{L}_j = R_j L_j/\pi_j$, $\widehat{H}_{j\ell} = R_j R_\ell H_{j\ell}/(\pi_j \pi_\ell)$, and $s_j = \widehat{L}_j + \sum_{\ell \neq j} \widehat{H}_{j\ell}$. Equation (12) reduces to

$$\widehat{V} = \sum_j (1-\pi_j) s_j s_j^\mathsf{T} - \sum_{j<\ell} (1-\pi_j)(1-\pi_\ell) \widehat{H}_{j\ell} \widehat{H}_{j\ell}^\mathsf{T}.$$

The first term counts a pair's diagonal covariance twice, once through each member. The subtraction supplies the required correction. The expression is unbiased but need not be positive semidefinite in every audit.

This decomposition also makes a normal approximation's requirements concrete. For a coefficient gradient g, the linear audit projection has variance $s^2 = \sum_j t_j (g^\mathsf{T} \nu_j)^2$. A sufficient Lindeberg condition is $\max_j |g^\mathsf{T} \nu_j|/(\pi_j s) \to 0$. The pair remainder must have variance $\sum_{j<\ell} t_j t_\ell (g^\mathsf{T} H_{j\ell})^2 = o(s^2)$. Variance-estimator consistency and a negligible Taylor remainder for the regression map are additional requirements. These conditions concern a sequence of panels and audits; they are not established merely by having many downstream records.

G Bias and the denominators of error rates

Let α_j be the probability of predicting one for a truly negative domain and β_j the probability of predicting zero for a truly positive domain, conditional on the browsing design. Then

$$b_j = \alpha_j(1-y_j) - \beta_j y_j, \qquad \mathrm{E}[d_i \mid C, y] = \sum_j c_{ij}\{\alpha_j(1-y_j) - \beta_j y_j\}. \tag{G1}$$

False-positive rates condition on true negatives; false-negative rates condition on true positives. Multiplying a false-positive rate by predicted positives uses the wrong denominator. Such a calculation would instead require the probability of a true negative conditional on a positive prediction, together with assumptions that justify transferring that probability to the weighted population. These conditional probabilities are not interchangeable.

If $\alpha_j = \alpha$ on negative domains and $\beta_j = \beta$ on positive domains,

$$\mathrm{E}[d_i \mid C, y] = \alpha(V_i - T_i) - \beta T_i = V_i\{\alpha - (\alpha+\beta)\pi_i\}, \qquad \pi_i = T_i/V_i, \tag{G2}$$

where the share π_i is defined for $V_i > 0$. At a fixed true share, expected bias scales with browsing volume unless the expression in braces is zero. For $\alpha + \beta > 0$, cancellation occurs at $\pi_i = \alpha/(\alpha+\beta)$. Equal positive error rates cancel only when half of visits are to truly positive domains. Thus the difference between FPR and FNR alone does not determine bias. Nor does the number of distinct visited domains: repeated visits matter through c_{ij}.

The common-rate assumption is stronger than knowing an average error rate on a validation set of domains. Errors may depend on popularity, domain type, or features associated with a particular group's browsing. In that case, equation (G1), with rates conditional on those features, is the relevant expression. The bias in a share is $\mathrm{E}[d_i \mid C, y]/V_i$; dividing by volume removes the mechanical scale factor, not misclassification bias.

With known α, β and $\alpha + \beta \neq 1$, this model gives an unbiased correction:

$$\widetilde{T}_i = \frac{\widehat{T}_i - \alpha V_i}{1 - \alpha - \beta}, \qquad \mathrm{E}[\widetilde{T}_i \mid C, y] = T_i. \tag{G3}$$

The correction is unstable when the denominator is close to zero and may return values outside $[0, V_i]$. Clipping changes its expectation. Estimating error rates introduces additional uncertainty shared across users; rates estimated on unweighted domains need not transfer to weighted exposure. This is the connection to classifier-based quantification (Vaz et al., 2019), with the additional requirement that the rates apply to each user's weights.

H A connection to label sampling

The same mapping supplies a remedy when a researcher can verify a probability sample of items. The companion `fewlab` project develops sampling designs for reusable labels (Sood, 2026). In its browsing application, the regression decomposition is $H = AC$, with column h_j giving the influence of domain j on the coefficient. For exposure shares, replace C by its row-normalized version. For any scalar target in equation (2), the corresponding item influence is w_j.

Freeze predictions q_j before drawing a label sample. Let R_j indicate selection, with known inclusion probability $s_j > 0$ for every item. Conditional on the fixed population and predictions, the difference estimator is

$$\widetilde{\theta} = \sum_j w_j q_j + \sum_j \frac{R_j}{s_j} w_j (y_j - q_j). \tag{H1}$$

Since $\mathrm{E}_R[R_j/s_j] = 1$, its expectation over label sampling is $\sum_j w_j y_j = \theta$, regardless of the classifier's errors. Verified labels must measure y_j correctly. Retraining the proxy on the same sampled labels requires a separate argument; the fixed-proxy proof does not cover it.

Under independent Bernoulli item sampling,

$$\mathrm{Var}_R(\widetilde{\theta}) = \sum_j \frac{1 - s_j}{s_j} w_j^2 (y_j - q_j)^2. \tag{H2}$$

The realized sample gives an unbiased variance estimator by replacing each summand with $R_j(1 - s_j) w_j^2 (y_j - q_j)^2 / s_j^2$. Fixed-size or balanced designs require their joint-inclusion terms or a justified approximation. An inaccurate proxy can increase variance relative to omitting it, but does not introduce design bias under these assumptions. This is uncertainty from randomized label acquisition, with y and q fixed; it is distinct from the prediction-error process in equation (J4). Both calculations preserve the reuse of an item label across downstream records.

I Auditing an aggregated regression covariate

The general plug-in error is $F(Cq, Z) - F(Cy, Z)$. If scalar F is differentiable throughout the segment from Cy to Cq, it also has the exact representation

$$F(Cq, Z) - F(Cy, Z) = \int_0^1 \left[C^\mathsf{T} \nabla_T F\{Cy + sC(q - y), Z\} \right]^\mathsf{T} (q - y)\, ds. \tag{I1}$$

Unlike the fixed weights of a linear target, these effective item weights depend on unknown labels along the path. Thresholds need not be differentiable.

Equation (7) gives the exact coefficient error. The following expansions connect it to the regression score and audit moments.

The conventional OLS score makes the two channels explicit. Write $a = C^\mathsf{T} M_V Y$, $G = C^\mathsf{T} M_V C$, and $U(z; \beta) = z^\mathsf{T} a - \beta z^\mathsf{T} G z$. For the fixed-sample complete-label coefficient $\beta^* = \beta(y)$, $U(y; \beta^*) = 0$, whereas

$$U(q; \beta^*) = e^\mathsf{T}(a - 2\beta^* G y) - \beta^* e^\mathsf{T} G e, \qquad e = q - y. \tag{I2}$$

If repeated predictions are meaningful while y, C, V, Y remain fixed, let $b = \mathrm{E}[e]$ and $\Sigma = \mathrm{Cov}(e)$. Then

$$\mathrm{E}[U(q;\beta^*)] = b^\mathsf{T}(a - 2\beta^* Gy) - \beta^*\{b^\mathsf{T}Gb + \mathrm{tr}(G\Sigma)\}. \tag{I3}$$

The trace term equals $\mathrm{tr}(M_V C \Sigma C^\mathsf{T})$. With independent item errors it is $\sum_j v_j \|M_V C_{\cdot j}\|^2$: each item's contribution depends on variation in its unit-level loading after removing the controls. It is nonnegative, but the other terms in equation (I3) can have either sign, and an expected score is not the expectation of the OLS coefficient. In particular, ordinary attenuation does not follow. For fixed binary y and hard binary q, nontrivial errors cannot have $b = 0$ item by item: a negative item can only have a positive error, and a positive item can only have a negative one. Mean-zero error is thus not a harmless default for this prediction process.

For several covariates built from the same predictions, let $T_r = C^{(r)}y$ and $d_r = C^{(r)}e$ on the same regression rows. The (r, s) error cross-product in their residualized normal equations has expectation

$$\mathrm{E}[d_r^\mathsf{T} M_V d_s] = b^\mathsf{T} C^{(r)\mathsf{T}} M_V C^{(s)} b + \mathrm{tr}\{C^{(r)\mathsf{T}} M_V C^{(s)} \Sigma\}. \tag{I4}$$

Even if distinct item errors are independent, the trace term contains $\sum_j v_j (M_V C^{(r)}_{\cdot j})^\mathsf{T} (M_V C^{(s)}_{\cdot j})$. One item prediction can therefore distort the cross-product between two constructed regressors. If the outcome also uses these predictions, its measurement error contributes the analogous term $\mathrm{E}[d_r^\mathsf{T} M_V d_0]$ to the regression numerator, with $C^{(0)}$ building the outcome. Shared errors can then create an apparent association as well as distort a regressor's denominator. These terms depend on how each aggregation weights the same item; they need not produce scalar attenuation of any coefficient.

More generally, let $D(y)$ collect p columns of the form $C^{(r)}y$. With an observed outcome Y, the complete-label coefficient vector is

$$\boldsymbol{\beta}(y) = B(y)^{-1} m(y), \qquad m_r(y) = y^\mathsf{T} C^{(r)\mathsf{T}} M_V Y, \qquad B_{rs}(y) = y^\mathsf{T} C^{(r)\mathsf{T}} M_V C^{(s)} y, \tag{I5}$$

when $B(y)$ is nonsingular. The same item label enters every component of m and B, so their audit uncertainty is joint. If Y is also built from these labels, its numerator moments are quadratic too. This is the usual regression normal equation with its item-level construction made explicit; it does not prescribe one particular correction method.

The fixed-sample coefficient in equation (6) has a linear numerator and a quadratic denominator in the item labels. Put $r = y - q$. Then

$$\begin{aligned} N &= y^\mathsf{T} a = q^\mathsf{T} a + r^\mathsf{T} a, \\ D &= y^\mathsf{T} G y = q^\mathsf{T} G q + 2 r^\mathsf{T} G q + r^\mathsf{T} G r. \end{aligned} \tag{I6}$$

Let R_j indicate an item's inclusion in a probability audit, with $\pi_j = \Pr_R(R_j = 1) > 0$ and $\pi_{j\ell} = \Pr_R(R_j = R_\ell = 1) > 0$ for every relevant pair. Set $\pi_{jj} = \pi_j$. The following estimates are design unbiased for the two moments, conditional on C, V, Y, y, q:

$$\begin{aligned} \widehat{N} &= q^\mathsf{T} a + \sum_j \frac{R_j}{\pi_j} r_j a_j, \\ \widehat{D} &= q^\mathsf{T} G q + 2 \sum_j \frac{R_j}{\pi_j} r_j (Gq)_j + \sum_{j,\ell} \frac{R_j R_\ell}{\pi_{j\ell}} r_j G_{j\ell} r_\ell. \end{aligned} \tag{I7}$$

The identities follow from $\mathrm{E}_R[R_j/\pi_j] = 1$ and $\mathrm{E}_R[R_j R_\ell/\pi_{j\ell}] = 1$. No independent-item-error or calibrated-score assumption is used. The ratio $\widehat{N}/\widehat{D}$ is generally not design unbiased and can be

unstable or undefined when the estimated denominator is near zero or nonpositive. Valid inference for the ratio requires additional design analysis; its variance involves inclusion probabilities for sets of up to four items. The formulas identify the information an item audit would have to preserve. The linear-label difference estimator in equation (H1) does not by itself correct a regression that uses Cy as a covariate.

For a retrospective standard error, treat $(\widehat{N}, \widehat{D})$ as a joint vector of item- and item-pair Horvitz–Thompson totals. If the required inclusion probabilities for sets of up to four items are known and positive, its design covariance can be estimated from the realized audit. When D is bounded away from zero and a normal approximation for the joint moments is justified, the delta-method variance for N/D uses gradient $(1/D, -N/D^2)^\mathsf{T}$. Near a zero denominator, a ratio confidence set must instead respect the joint uncertainty in numerator and denominator; the plug-in delta standard error is unreliable. With several constructed regressors, the corresponding moment vector includes all residualized regressor–outcome and regressor–regressor cross-products, including equation (I4). Its covariance and the regression map must be propagated jointly. These are routes to inference under an observed probability-validation design, not a claim that an arbitrary validation set supports an automatic corrected standard error.

J Repeated-prediction error and dependence

For a fixed fitted classifier and fixed labels, Δ is fixed. To discuss bias and variance, specify a repeated-prediction process, such as repeated training samples, while fixing y, w and the inputs that determine w. All moments below condition on this fixed design. Define $b = \mathrm{E}[e]$ and the full joint covariance $\Sigma = \mathrm{Cov}(e)$, without assuming independence across items. Then

$$\begin{aligned} \mathrm{Bias}(\widehat{\theta}) &= w^\mathsf{T} b, \\ \mathrm{Var}(\Delta) &= w^\mathsf{T}\Sigma w = \sum_j w_j^2 \Sigma_{jj} + 2\sum_{j<\ell} w_j w_\ell \Sigma_{j\ell}, \qquad \text{(J1)} \\ \mathrm{E}[\Delta^2] &= (w^\mathsf{T} b)^2 + w^\mathsf{T}\Sigma w. \qquad \text{(J2)} \end{aligned}$$

Equations (1)–(J2) connect the research quantity to its item weights, systematic error, and uncertainty. Covariance between item errors enters the estimate's variance. Dependence alone does not determine bias; systematic errors on items with nonzero weights can make $w^\mathsf{T} b$ nonzero.

Unit aggregates and regression vectors are corresponding joint quantities. Writing $\widehat{T} = Cq$, $d = \widehat{T} - T$, and $\widehat{\beta} = ACq$ gives

$$d = Ce, \qquad \text{(J3)}$$

$$\mathrm{E}[d] = Cb, \qquad \mathrm{Cov}(d) = C\Sigma C^\mathsf{T}, \qquad \text{(J4)}$$

$$\mathrm{E}[\widehat{\beta} - \beta^*] = ACb, \qquad \mathrm{Cov}(\widehat{\beta} - \beta^*) = AC\Sigma C^\mathsf{T} A^\mathsf{T}. \qquad \text{(J5)}$$

The same item error is reused across every unit linked to that item. These formulas quantify the specified prediction-error process; they do not include errors in the item-to-unit map, unit sampling, or uncertainty about the construct. They concern a label-derived aggregate as an outcome. Using that aggregate as a regressor is a different errors-in-variables problem treated in Section 3.

J.1 Shared labels and correlated item errors

These are separate mechanisms under a repeated-prediction model. First, a single item error can appear in several units' aggregates. Even if the item errors e_j are independent across j, units that

share an uncertain item can have correlated aggregate errors. This is a standard common-shock structure, analogous to shift-share inference (Adão et al., 2019); it does not assert correlation between distinct item errors. Second, distinct item errors may themselves covary under repeated fitting. Shared training data or model parameters can produce such covariance, but using one fitted classifier does not by itself establish it. Units linked to disjoint items can have correlated aggregate errors only under this second mechanism or another shared source of uncertainty. In equation (J1), positively correlated item errors increase variance when their weights have the same sign, and decrease it when their weights have opposite signs.

For a limiting example, suppose centered errors are shared within item families: $e_j - b_j = u_{g(j)}$, with independent family shocks u_g of variance τ_g^2. This is compatible with binary predictions when the classifier assigns one common random prediction to every item in a family. Then

$$\mathrm{Var}(\Delta) = \sum_g \tau_g^2 \left(\sum_{j:g(j)=g} w_j \right)^2 . \tag{J6}$$

A family of m items with unit weights contributes $m^2\tau_g^2$, whereas assuming independent item errors would give $m\tau_g^2$. Signed weights can instead cancel within a family. The relevant concentration is in aggregate family influence, not just individual item influence. This example is a special covariance structure, not an assertion that the right families are known or that all within-family mistakes are perfectly correlated.

Independent item errors are another special case: Σ is diagonal, with entries v_j. For any unit aggregate $T = Cy$, equation (J4) reduces to

$$\mathrm{Var}(d_i) = \sum_j c_{ij}^2 v_j, \qquad \mathrm{Cov}(d_i, d_\ell) = \sum_j c_{ij} c_{\ell j} v_j. \tag{J7}$$

Let α_j be the false-positive probability on a truly negative item and β_j the false-negative probability on a truly positive item. For binary predictions, $v_j = \alpha_j(1-\alpha_j)$ on negative items and $v_j = \beta_j(1-\beta_j)$ on positive items. In browsing, repeated visits multiply one classification error; treating visits as independent would incorrectly replace squared visit weights by visit counts. More generally, any reuse of an uncertain item can induce dependence across units.

The same predictions can build several different aggregates. If the rth aggregate uses a fixed matrix $C^{(r)}$, then

$$d^{(r)} = C^{(r)} e, \qquad \mathrm{Cov}\{d^{(r)}, d^{(s)}\} = C^{(r)} \Sigma C^{(s)\mathsf{T}}. \tag{J8}$$

For scalar estimates with item-weight vectors $w^{(r)}$, independent item errors yield $\mathrm{Cov}(\Delta_r, \Delta_s) = \sum_j v_j w_j^{(r)} w_j^{(s)}$. Thus each item contributes a rank-one term to the joint covariance of all estimates. A heavily weighted item can move several reported quantities together; subtracting two of them can cancel its error. For a sequence of larger unit samples, independent item errors with $v_j\{w_j^{(r)}\}^2$ bounded away from zero for even one item prevent the prediction-error variance of that estimate from vanishing. More units alone do not supply more independent predictions for that item. The covariance statements require a repeated-prediction model; $d^{(r)} = C^{(r)}e$ remains an exact identity for fixed realized predictions.

For a fixed linear unit contrast $a^\mathsf{T} d$, equal item-error variances give a simple overlap diagnostic:

$$R(a, C) = \frac{\sum_j (\sum_i a_i c_{ij})^2}{\sum_i a_i^2 \sum_j c_{ij}^2}. \tag{J9}$$

The numerator is its variance, apart from the common variance factor. The denominator is what that variance would be if each unit's aggregate error were independent while retaining its marginal

variance. Thus R measures the effect of discarding covariance across units for this one prediction-error component. It is at least one for nonnegative weights and counts, but may be above or below one for a signed contrast. It is not a general standard-error correction: unequal item variances, correlated item errors, unit sampling, and other errors require their own treatment.

An uncertainty procedure must follow the same structure as the estimand. Resampling downstream records while holding predictions fixed does not propagate repeated-training uncertainty. Each replicate must preserve reused labels across records and the joint distribution of errors across items. Independently perturbing each item label imposes a diagonal Σ; resampling the classifier fit and predicting all items together can preserve dependence induced by that fitting process, provided the resampling scheme represents that process. A prediction-error covariance cannot simply be added to an arbitrary reported regression covariance: the components may overlap or be dependent. Neither larger downstream samples nor clustering records alone guarantees that this error disappears, and standard errors do not repair bias.

J.2 Many items do not ensure a normal approximation

Even with independent item errors, diffuse variance shares alone do not justify Gaussian inference. Let all true labels be zero and let each of k items be falsely labeled positive independently with probability $1/k$. With $w_j = 1$, total error B_k is binomial with mean one and variance $1 - 1/k$. The largest item variance share is $1/k$, yet $B_k - 1 \Rightarrow \text{Poisson}(1) - 1$, not a normal distribution. At $k = 10{,}000$, an oracle interval using the exact bias and variance but a normal 1.96 cutoff has coverage 91.97%. This calculation uses exact binomial probabilities, not simulation.

For independent bounded binary errors, a sufficient Lindeberg condition is

$$\frac{\max_j |w_j|}{\sqrt{\sum_j w_j^2 v_j}} \to 0.$$

A small maximum variance share $\max_j w_j^2 v_j / \sum_j w_j^2 v_j$ is weaker when errors become rare. The example violates the former condition and satisfies the latter. A growing number of items cannot substitute for a condition on the distribution of their errors. This is a prediction-error example, distinct from the fixed-label audit design.

K Count-certificate construction and audit diagnostics

Partition items into strata $h = 1, \ldots, H$ before observing their audited labels. Sample n_h of the M_h items without replacement. If K_h is the unknown number of incorrect predictions and X_h the sampled error count, then $X_h \sim \text{Hypergeometric}(M_h, K_h, n_h)$. Assign nonnegative error budgets δ_h with $\sum_h \delta_h \le \delta$. Retain integers K for which both $\Pr_K(X_h \le x_h) \ge \delta_h/2$ and $\Pr_K(X_h \ge x_h) \ge \delta_h/2$. Their extrema are ℓ_h, u_h. A census stratum has $\ell_h = u_h = x_h$; an unaudited stratum contributes no count restriction.

Let O be the audited items, U_h the unaudited items in stratum h, $c_j = (2q_j - 1)w_j$, and $d_O = \sum_{j \in O} w_j(q_j - y_j)$. The smallest and largest compatible errors are

$$\begin{aligned} L_A &= d_O + \sum_h \min_{\ell_h - x_h \le m \le u_h - x_h} S_-(c_{U_h}, m), \\ U_A &= d_O + \sum_h \max_{\ell_h - x_h \le m \le u_h - x_h} S_+(c_{U_h}, m). \end{aligned} \tag{K1}$$

The optimization is over integers. With signed weights, an extreme need not occur at the largest admissible error count.

Proof of Proposition 4. Inverting each pair of hypergeometric tails excludes the true K_h with probability at most δ_h. A union bound therefore includes all true counts with probability at least $1-\delta$. On that event, the true errors in U_h number between $\ell_h - x_h$ and $u_h - x_h$. Sorting and choosing a count in that range gives the extrema in equation (K1); the choices are independent restrictions across strata and hence jointly attainable. The true label vector lies in the same configuration set for every w, so the coverage event applies simultaneously to all its linear contrasts. □

The labels and predictions are fixed under this design. Systematic or correlated classifier mistakes do not invalidate coverage, which is over repeated audit samples. The count region can be conservative, and sharp projection of that region does not make its interval shortest. The guarantee concerns one label vector; combining separate audits of several outcomes needs an additional error allocation. Optional stopping, adaptive strata, and imperfect audit labels require other procedures or assumptions.

Information lost by count projection. Let all N predictions equal one, half the weights equal $1/N$ and half equal $-1/N$. Suppose a fraction $p < 1/2$ of labels in each half are wrong. The true contrast is zero. If $n \to \infty$ but $n/N \to 0$, an unstratified audit consistently learns the overall error fraction. Yet unaudited errors can still be placed almost entirely on either weight sign: the projected interval width converges to $2p$. A direct estimate of the weighted audit residual instead has standard error of order $n^{-1/2}$. Separating the two weight signs into strata removes this particular information loss.

For a declared contrast, the stratified difference estimator uses $z_j = w_j(y_j - q_j)$ directly:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j \in O_h} z_j, \qquad \widehat{V} = \sum_h \frac{M_h^2(1 - n_h/M_h)}{n_h} s_{z,h}^2. \tag{K2}$$

Under independent stratum draws, the estimate and variance estimator are design unbiased when each noncensus stratum has at least two sampled items. Normal intervals need a separate approximation. A conservative finite-sample alternative applies Hoeffding's without-replacement inequality (Hoeffding, 1963) to each bounded residual mean and sums the stratum radii. More efficient weighted audit procedures exist (Shekhar et al., 2023).

The random and half-certainty designs were chosen for one coefficient. They do not establish that this design is efficient for other contrasts. The count certificate covered the target in every simulated draw but never excluded zero. Direct normal intervals were much narrower, yet coverage in the smallest random design fell below the nominal level. The finite-sample Hoeffding baseline was wider than count projection. These comparisons do not identify a finite-sample-valid, efficient direct interval for this design.

The narrower exact-count range from finer strata need not translate into a narrower audit interval: the fixed sample is divided among more strata and the error budget is split across their discrete hypergeometric intervals. The 10- versus 50-stratum mean-width difference should be read with the reported Monte Carlo uncertainty, not as evidence of an optimal partition.

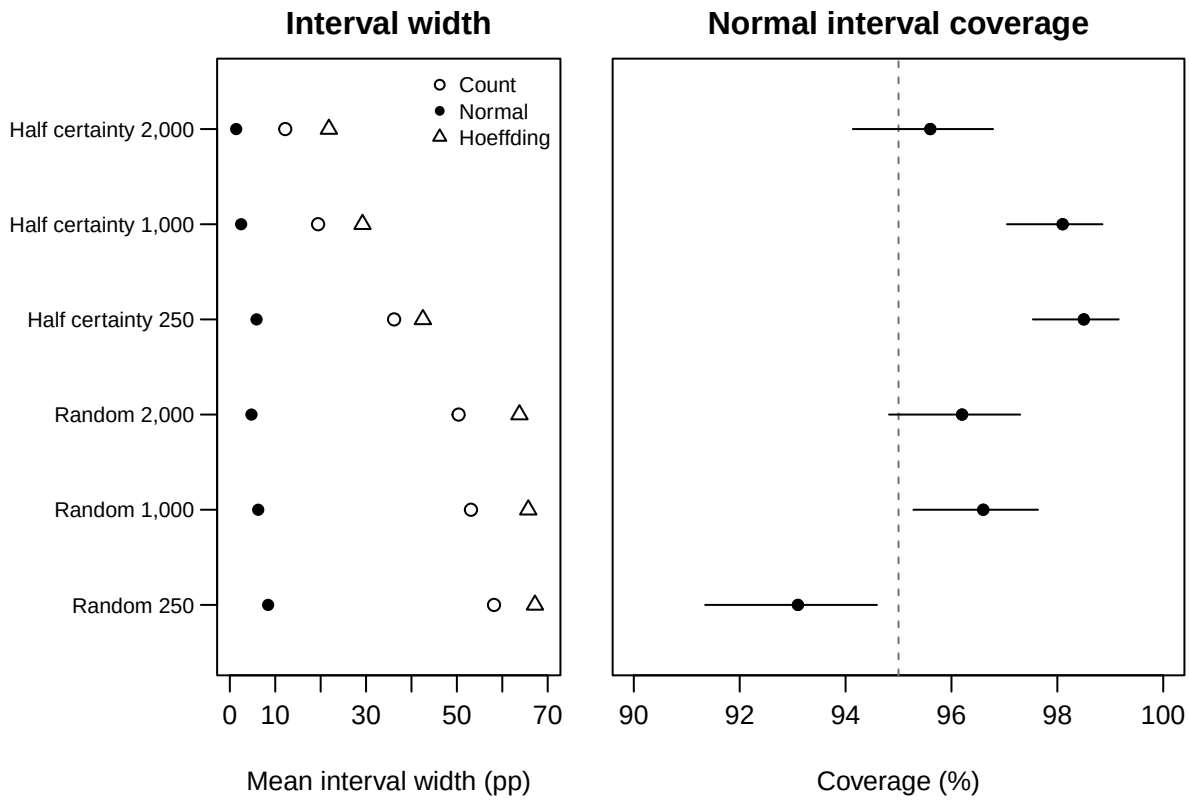


Figure K1: Width and coverage for the Facebook-pixel female share coefficient. Rows compare audit budgets and designs in the fixed browsing frame. Left: mean interval width in percentage points over 1,000 audits. Right: normal interval coverage, with exact 95% binomial Monte Carlo intervals and a dashed 95% reference. Count and Hoeffding intervals covered in every simulated draw. "Half certainty" audits the largest absolute influences with half the budget. Count intervals cover all contrasts of this outcome simultaneously; direct intervals target this coefficient. The randomness is domain auditing; users and predictions are fixed.

Table K1: All audit designs and methods for the focal coefficient

Design	Budget	Interval	Coverage (%)	Width (pp; MCSE)	Correct sign (%)	Wrong sign (%)
Random	250	Count projection	100.0	58.2 (0.06)	0.0	0.0
Random	250	Direct normal	93.1	8.4 (0.50)	39.6	0.0
Random	250	Direct Hoeffding	100.0	67.2 (0.01)	0.0	0.0
Random	1,000	Count projection	100.0	53.1 (0.05)	0.0	0.0
Random	1,000	Direct normal	96.6	6.3 (0.19)	32.9	0.0
Random	1,000	Direct Hoeffding	100.0	65.7 (0.02)	0.0	0.0
Random	2,000	Count projection	100.0	50.4 (0.05)	0.0	0.0
Random	2,000	Direct normal	96.2	4.8 (0.11)	34.4	0.0
Random	2,000	Direct Hoeffding	100.0	63.8 (0.03)	0.0	0.0
Half certainty	250	Count projection	100.0	36.2 (0.05)	0.0	0.0
Half certainty	250	Direct normal	98.5	5.9 (0.29)	40.6	0.0
Half certainty	250	Direct Hoeffding	100.0	42.5 (0.00)	0.0	0.0
Half certainty	1,000	Count projection	100.0	19.4 (0.04)	0.0	0.0
Half certainty	1,000	Direct normal	98.1	2.5 (0.04)	41.9	0.0
Half certainty	1,000	Direct Hoeffding	100.0	29.2 (0.00)	0.0	0.0
Half certainty	2,000	Count projection	100.0	12.2 (0.03)	0.0	0.0
Half certainty	2,000	Direct normal	95.6	1.4 (0.01)	65.0	0.0
Half certainty	2,000	Direct Hoeffding	100.0	21.8 (0.00)	0.0	0.0

Each row summarizes 1,000 domain audits. Width is the mean in percentage points, with its Monte Carlo standard error in parentheses. Sign columns report exclusion of zero in the correct or wrong direction. All methods within a design use the same draws. Exact binomial Monte Carlo intervals and replicate results accompany the package.

Table K2: Refining validation strata for the focal coefficient

Strata	Exact-count width (pp)	Audit width (pp; MCSE)	Coverage (%)
2	49.7	53.2 (0.05)	100.0
10	34.7	47.1 (0.05)	100.0
50	21.8	48.2 (0.03)	100.0

Rank bins of signed influence cross predicted labels. Exact-count width conditions on the population count in every stratum; audit width averages 95% count-projection intervals over 1,000 domain samples of size 1,000. Width uncertainty is its Monte Carlo standard error. These partitions were examined after the initial audit results. No design certified the sign.

L Retaining the highest-visit domain labels

Table L1 repeats the count-information exercise after retaining both observed classification labels for the highest-visit domains. For the remaining domains it retains only the numbers positive under Piedomains alone and under the provider alone. The ordering uses visit count with domain name to break ties. The count range is sharp under exactly this information: the head contribution is fixed, while the tail discrepancies can occupy any compatible tail domains. At zero retained domains it equals the range in Table 3. The provider's observed visit share lies inside every range.

Table L1: Count-compatible provider visit shares after retaining label pairs for the highest-visit domains

Provider	Domains retained	Visits retained (%)	Count range (%)	Width (pp)
Bitdefender	0	0.0	[0.07, 82.11]	82.05
Bitdefender	10	42.2	[0.07, 40.06]	39.99
Bitdefender	50	56.8	[0.07, 25.80]	25.73
Bitdefender	250	71.5	[1.00, 13.65]	12.64
Bitdefender	1,000	83.4	[1.33, 5.98]	4.64
Forcepoint	0	0.0	[0.07, 80.61]	80.54
Forcepoint	10	40.6	[0.07, 40.16]	40.09
Forcepoint	50	55.1	[0.07, 25.99]	25.92
Forcepoint	250	69.6	[0.86, 14.01]	13.15
Forcepoint	1,000	81.9	[1.24, 6.24]	4.99
alphaMountain	0	0.0	[0.04, 68.07]	68.03
alphaMountain	10	30.1	[0.04, 38.33]	38.30
alphaMountain	50	49.0	[0.56, 21.37]	20.82
alphaMountain	250	67.2	[1.44, 9.12]	7.68
alphaMountain	1,000	81.5	[1.87, 4.03]	2.16

The percentages use each provider row's categorized-domain visit total as its denominator. "Visits retained" is the share of that total on domains whose two observed labels remain known. Width is in percentage points. The providers use different category definitions and neither label is an adjudicated content truth. Retaining a label pair here is an information exercise, not a claim that a manual audit was performed.